

Lab 11 – Implement Monitoring (AZ-104)

Full Notes + README + Repo Structure + Final AZ-104 README

LAB 11 – FULL DETAILED NOTES

◆ What this lab is about (Big Picture)

Azure Monitor helps you:

- **Observe** what is happening in Azure
- **Detect issues early**
- **Alert admins automatically**
- **Analyze logs and metrics**

In real projects, **monitoring is mandatory**, not optional.

Core Azure Monitoring Components

Component	Purpose
Azure Monitor	Central monitoring platform
Metrics	Numeric values (CPU %, memory)
Logs	Detailed records (events, heartbeats)
Alerts	Notify when conditions are met
Action Groups	Who gets notified
Log Analytics	Query and analyze logs
KQL	Query language for logs

TASK 1 – Deploy VM using ARM Template

Why?

We need a **resource to monitor**.

What happens:

- ARM template deploys:
 - 1 Virtual Network
 - 1 Virtual Machine
- Region: East US
- VM is later used for alerts, logs, heartbeats

Key Learning

- Monitoring works **on resources**
 - ARM ensures **consistent deployment**
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TASK 1.1 – Enable VM Insights

What is VM Insights?

- Collects:
 - CPU, Memory
 - Performance counters
 - Heartbeat signals
- Installs **Azure Monitor Agent**

Why it matters:

Without Insights → **no logs, no metrics**

TASK 2 – Create Alert Rule (VM Deletion)

Alert Type

Activity Log Alert

Triggered when:

- Virtual Machine is **deleted**

Why Activity Logs?

- Track **management operations**
- Who did what, when, and where

Key Fields:

- Signal: Delete Virtual Machine
 - Scope: Subscription
 - Event Level: All
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 TASK 3 – Configure Action Group**What is an Action Group?**

A notification channel.

Supports:

- Email
- SMS
- Push notifications
- Webhooks

What we configured:

- Email notification to admin
- Reusable across alerts

Interview Tip:

One action group can be reused by multiple alert rules.

 TASK 4 – Trigger & Verify Alert**How we tested:**

- Deleted VM
- Alert triggered automatically
- Email notification received

Important:

Alerts may take **1–5 minutes** to appear.

 TASK 5 – Alert Processing Rule

What is it?

Controls **WHEN alerts notify**, not WHEN they trigger.

Use case:

- Maintenance windows
- Planned downtime

Configuration:

- Suppress notifications
- Time-based (10 PM – 7 AM)

Difference:

Alert Rule	Alert Processing Rule
Detects issue	Controls notification
Always evaluates	Can mute alerts



TASK 6 – Log Analytics & KQL Queries

Heartbeat Query

Heartbeat

```
| summarize count() by Computer
```

What does count = 5 mean?

- ✓ VM sent **5 heartbeat signals**
- ✓ Agent is working
- ✓ VM was alive during that period

CPU Utilization Query

InsightsMetrics

```
| where TimeGenerated > ago(1h)
```

```
| where Name == "UtilizationPercentage"
```

```
| summarize avg(Val) by bin(TimeGenerated, 5m), Computer
```

```
| render timechart
```

Why this matters:

- Visual performance monitoring
 - Can convert query → Alert
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Cleanup (Important)

- Delete resource group
 - Avoid unnecessary costs
 - Alerts & logs stop automatically
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LAB 11 KEY TAKEAWAYS

- Azure Monitor is **central observability platform**
 - Alerts + Action Groups automate responses
 - Logs give **deep insights**
 - KQL is critical for troubleshooting
 - Alert Processing Rules prevent alert fatigue
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LAB 11 – INTERVIEW Q&A

Q1. Difference between Metrics and Logs?

Metrics	Logs
Numeric	Text-based
Near real-time	Detailed
Lightweight	Heavy data

Q2. What triggers Activity Log alerts?

- Azure management operations (Create, Delete, Update)
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Q3. What is an Action Group?

A reusable notification object for alerts.

Q4. What is Heartbeat?

Signal sent by VM agent to confirm VM health.

Q5. What is KQL?

Kusto Query Language used to query Azure Monitor logs.